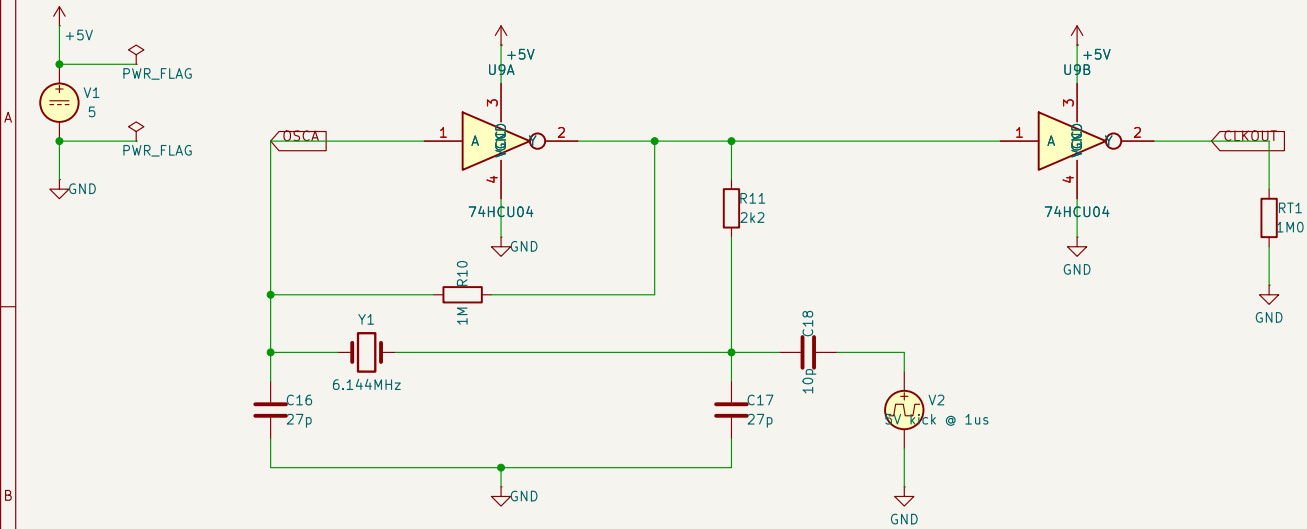


I Pierce oscillator (74HCU04 + 6.144 MHz crystal -- the 2-pin part that replaced the unobtainable can)



The board's exact topology: Rf 1M biases the gate linear, Rs 2k2 + C17 damp the crystal drive -- and kill the ~30 MHz parasitic through Co that Rs=1k lets a gate sustain. UNBUFFERED 74HCU04 only; a buffered 74HC04's three-stage gain re-lights that parasitic at any Rs. And the second gate squares the raw node up for the jumper. The crystal model resonates SERIES at exactly 6.144 MHz; parallel-mode operation pulls +Cm/2(CL+Co) ~ +0.05 %, so the frequency check brackets that, not fs itself. Startup rides a 5 V kick through 10 p at t=1u; what is asserted is that the loop SUSTAINS it at full swing long after the kick is gone.

.tran 2n 400u uic

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